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# Evaluating Pedagogical Structure in AI-Generated Introductory Physics Explanations

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## Abstract

1 AI systems often answer introductory physics questions correctly, yet students  
2 need explanations that expose the reasoning behind the answer. We study whether  
3 AI-generated explanations contain pedagogical discourse moves such as framing  
4 the physical situation, applying principles, checking conclusions, giving intuition-  
5 building examples, and marking caveats. This framing also matters for epistemic  
6 safety in AI tutoring: fluent incomplete explanations can invite misplaced learner  
7 confidence. Using 60 explanations across 20 introductory physics prompts, we  
8 compare human-written and AI-generated responses with sentence-level labels,  
9 quality ratings, and a lightweight keyword baseline. The annotation scheme reaches  
10 Cohen’s  $\kappa = 0.6706$  on a 72-pair overlap subset. In this pilot, AI explanations  
11 match human explanations on correctness, show lower intuition-bridging coverage  
12 (20% vs 35%), and receive lower completeness ratings (4.0 vs 5.0). The findings  
13 suggest that AI physics tutor evaluation should measure explanation structure  
14 alongside answer correctness.

## 15 1 Introduction

16 Evaluation at the intersection of physics and AI has advanced quickly, especially for answer-level  
17 reasoning tasks. Educational settings place an additional demand on these systems. A useful physics  
18 tutor needs to frame concepts, make reasoning legible, and connect formal equations to learner  
19 intuition. In practice, many LLM outputs are fluent and locally plausible while still being incomplete  
20 as teaching artifacts.

21 Consider the question: why does an object moving in a circle accelerate even if its speed is constant?  
22 A thin answer may state that centripetal acceleration points inward. A stronger teaching explanation  
23 first addresses the misconception that constant speed implies no acceleration, explains that velocity  
24 includes direction, and then gives an intuitive example such as a car turning around a curve. Only the  
25 second exposes the reasoning a beginner needs to calibrate trust.

26 This distinction is also a small AI safety issue for educational deployment. When a response sounds  
27 coherent and authoritative, students may overtrust it even if key assumptions, caveats, or conceptual  
28 bridges are missing. We study this risk through explanation structure. The paper contributes:

- 29 • a compact 5-label discourse taxonomy with usable pilot reliability ( $\kappa = 0.6706$  on 72  
30 overlap pairs),
- 31 • a paired human and AI study showing lower AI intuition-bridging coverage at explanation  
32 level (-15 pp), and
- 33 • a lightweight keyword baseline for sentence labeling, included to anchor the benchmark as  
34 a reproducible evaluation task.

| Category                | Prompts | Human | AI |
|-------------------------|---------|-------|----|
| Mechanics               | 8       | 8     | 16 |
| Electricity & Magnetism | 8       | 8     | 16 |
| Thermo/Energy           | 4       | 4     | 8  |
| Total                   | 20      | 20    | 40 |

Table 1: Pilot dataset composition.

| Label     | Plain meaning               | Example function                                |
|-----------|-----------------------------|-------------------------------------------------|
| FRAME     | Sets up the situation       | Constant speed differs from constant velocity.  |
| PRINCIPLE | States or applies a rule    | Acceleration is change in velocity over time.   |
| VERIFY    | Checks a conclusion         | Without inward force, motion would be straight. |
| INTUITION | Gives an example or analogy | A car turning around a curve changes direction. |
| CAVEAT    | Marks a condition or limit  | This assumes air resistance is negligible.      |

Table 2: Plain-English label guide used to make the discourse scheme readable beyond NLP specialists.

## 2 Related Work

Recent benchmarks such as PhysReason and UGPhysics provide broad coverage of physics reasoning at answer level [Zhang et al., 2025, Xu et al., 2025]. The present study targets a narrower educational question: whether a correct explanation contains the discourse moves that support learning. Methodologically, the annotation protocol follows established reliability guidance in computational linguistics [Cohen, 1960, Landis and Koch, 1977, Artstein and Poesio, 2008], adapted to a compact pedagogy-oriented taxonomy. The contribution is a discourse-aware evaluation layer for introductory physics explanation quality.

## 3 Dataset and Annotation Protocol

### 3.1 Data Scope and Sources

The study uses 20 prompts and 60 explanations (1 human + 2 AI per prompt), distributed across mechanics (8 prompts), electricity and magnetism (8), and thermo/energy (4). Human references come from public introductory physics resources licensed for reuse and were normalized into a comparable explanatory format [OpenStax, 2026].

### 3.2 Annotation Schema

Each sentence receives exactly one of five labels (Table 2). During calibration, principle statements and direct applications were merged into a single PRINCIPLE category because annotators most often confused that boundary. Explanation-level ratings are 1–5 for clarity, correctness, and completeness.

### 3.3 Reliability Protocol

A 72-pair overlap subset was independently labeled by two annotators for calibration. Final agreement is  $\kappa = 0.6706$  and  $P_o = 0.8333$ , which is suitable for pilot benchmark analysis with transparent limitations.

### 3.4 Heuristic Baseline

As a simple baseline, a rule-based classifier assigns sentence labels from cue phrases, such as analogy markers for INTUITION and qualification markers for CAVEAT. On 381 singly labeled sentences, this baseline reaches 70.6% accuracy and 0.575 macro-F1. Most errors collapse VERIFY and CAVEAT into PRINCIPLE, giving a lower-bound comparison for future automated labeling work.

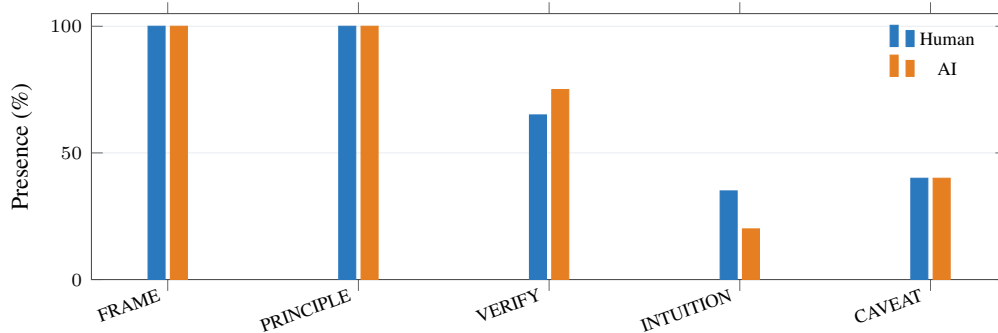


Figure 1: Explanation-level discourse move presence. AI responses show lower INTUITION prevalence despite matching human responses on FRAME and PRINCIPLE.

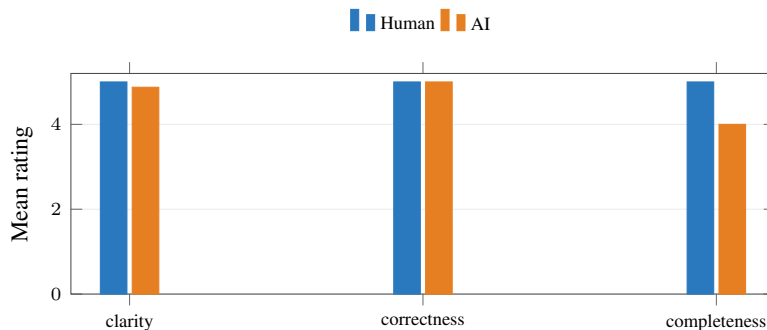


Figure 2: Holistic ratings by source. Correctness is saturated; completeness carries the strongest separation signal.

## 62 4 Results

### 63 4.1 Move Prevalence by Source

64 Figure 1 shows explanation-level move presence rates. Both sources are saturated on FRAME and  
 65 PRINCIPLE (100%), then diverge on lower-frequency pedagogical moves. The strongest gap is  
 66 INTUITION (35% human vs 20% AI), while AI shows higher VERIFY presence (75% vs 65%).

67 Sentence-level distributions are close on dominant structural content (PRINCIPLE: 55.37% human,  
 68 54.62% AI). The difference is concentrated in learner-facing bridges, with core inferential content  
 69 remaining similar. This matters because introductory students often struggle to map a formal statement  
 70 onto a physical situation. An explanation can be correct while still leaving that mapping implicit.

### 71 4.2 Holistic Ratings

72 Figure 2 and Table 3 summarize ratings from the primary annotator. Correctness is saturated (5.0 for  
 73 both), while completeness shows the largest separation (5.0 human vs 4.0 AI).

### 74 4.3 Uncertainty

75 For key move contrasts, uncertainty remains wide at pilot scale: INTUITION difference is  $-15.0$   
 76 pp (95% CI:  $[-39.3, 9.3]$ ,  $p = 0.206$ ) and VERIFY is  $+10.0$  pp (95% CI:  $[-14.8, 34.8]$ ,  $p = 0.418$ ).  
 77 Prompt-level leave-one-out checks preserve the negative INTUITION direction ( $[-21.1, -10.5]$  pp),  
 78 although multiple-comparison-adjusted significance is not reached. We report this as directional pilot  
 79 evidence. In qualitative review, the circular-motion examples show the same pattern: AI responses  
 80 identify inward acceleration, while the human explanation more directly repairs the constant-speed  
 81 misconception by distinguishing speed from velocity.

| Source             | Clarity | Correctness | Completeness |
|--------------------|---------|-------------|--------------|
| Human ( $n = 20$ ) | 5.000   | 5.000       | 5.000        |
| AI ( $n = 40$ )    | 4.875   | 5.000       | 4.000        |
| AI - Human         | -0.125  | 0.000       | -1.000       |

Table 3: Mean holistic ratings.

## 5 Implications and Limitations

The practical implication is that discourse labels can become a diagnostic layer for AI tutoring systems. A tutor response could be checked for correctness and then inspected for missing pedagogical moves: no physical framing, no everyday example, no stated idealization, or no misconception repair. A revision system could then ask the model to add the missing move. This does not establish learning gains. It does reduce one class of epistemic safety risk: fluent explanations that invite overtrust while leaving key reasoning steps hidden.

This is a pilot study. The dataset is small, interval estimates for move prevalence are wide, and aggregate ratings rely on one primary annotator. The human explanations were normalized for comparability, which reduces stylistic noise but may compress natural human variation. Completeness ratings also show no within-group variance in this summary, so inferential claims about that metric should remain cautious. The next version should use a larger multi-rater corpus, unrevised human sources, and learner-outcome validation.

## 6 Conclusion

This pilot study contributes a compact protocol for evaluating the pedagogical structure of AI-generated physics explanations. The evidence suggests that educational AI systems should be evaluated for correctness and for the reasoning structure that helps learners understand, question, and calibrate trust in an explanation.

## References

- Ron Artstein and Massimo Poesio. Inter-coder agreement for computational linguistics. *Computational Linguistics*, 34(4):555–596, 2008.
- Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960.
- J. Richard Landis and Gary G. Koch. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174, 1977.
- OpenStax. Openstax licensing and attribution information. <https://openstax.org/books/college-physics-2e/pages/6-2-centripetal-acceleration>, 2026.
- Xin Xu, Qiyun Xu, Tong Xiao, Tianhao Chen, Yuchen Yan, Jiaxin Zhang, Shizhe Diao, Can Yang, and Yang Wang. Ugphysics: A comprehensive benchmark for undergraduate physics reasoning with large language models, 2025. arXiv:2502.00334.
- Xinyu Zhang, Yuxuan Dong, Yanrui Wu, Jiaying Huang, Chengyou Jia, Basura Fernando, Mike Zheng Shou, Lingling Zhang, and Jun Liu. Physreason: A comprehensive benchmark towards physics-based reasoning, 2025. arXiv:2502.12054.